

Lazy Closed Walks in Growing Dimension

All fixed orders, parity conditioned Poisson laws,
and occupied coordinate fluctuations for OEIS A328716

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Abstract

Let a_n count the n -step walks from the origin to itself in $\mathbb{Z}^n$, with steps $0, \pm e_1, \dots, \pm e_n$. This is OEIS A328716. We prove its known parity-dependent leading equivalent, give exact Bessel definitions of its constants, and construct every fixed inverse-power correction, including three explicit terms. The marked expansion has an absolute complex-uniform remainder, so it remains valid when its leading amplitude vanishes.

For a uniformly chosen bridge, the zero-step count has an all-orders exponentially weighted ℓ^1 expansion about a parity-conditioned Poisson law. The number of occupied coordinate axes has Gaussian fluctuations, asymptotically independent of that residual Poisson variable, and an explicit parity-dependent constant mean correction. Separate inverse carriers retain both parity constants and give an eventual global integer bracket of width at most one. A compact proportional length-to-dimension extension retains the moving Poisson parameter.

The exact generating function and numerical leading equivalent were already recorded in the OEIS. Bessel walk enumeration and full large-powers expansions are classical. This is a family-specific derivation and refinement, with bounded source comparison, not a claim of an open-problem resolution or historical priority. Finite exact verification and optional high-precision diagnostics accompany the proofs; no effective inverse error constant or certified numerical starting threshold is claimed.

[write] **Status and trust boundary, added 7 October 2026, batch 109.** This is bundle Report 179 of an external AI-assisted research session, filed in ProveIt as the single-source research report **a328716-lazy-closed-walks** (provenance, the sources as the write read them, what was checked, relation to the repository, the collected non-claims and the reading conventions are in Section 1.1). It is unrefereed and not formalized. Its author line and PDF author field read “Research report 179”: it names no person or tool. *What rests on what.* Proposition 2.1, Lemma 3.1 and Theorems 4.1, 6.1, 7.1, 8.1 and 9.1 have conventional proofs in the text: the Bessel-product count, a two-arc saddle analysis with an absolute integrated remainder, Cauchy estimates on a larger disk, an analytic moving saddle with Lévy’s continuity theorem, and monotone parity brackets. The explicit $c_{\epsilon,2}$, $c_{\epsilon,3}$ come from the finite algorithm of Section 4, checked by the shipped programs; every decimal is a diagnostic. The exact formula and Kotěšovec’s numerical leading equivalent are in the OEIS, the Bessel walk generating function is Gessel–Weinstein–Wilf’s, and the large-powers method is classical (Flajolet–Sedgewick). The write adds Remarks 1.1, 8.2 and 9.2 (the OEIS entries quoted and the b-file compared; the inverses against the transseries volume; a proof of the row conjecture of A328718, a fixed-dimension statement outside the source’s regime) and dated notes. No statement, proof or number of the source was changed.

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1 The model and the scope of the results

A walk is an ordered sequence of steps from $\{0, +e_1, -e_1, \dots, +e_D, -e_D\}$, where the axes are labeled and the zero step is a single available step. A bridge has total displacement zero. Let

$$A_{N,D}(u) = \sum_{N\text{-step bridges in } \mathbb{Z}^D} u^{\text{number of zero steps}}, \quad A_n(u) = A_{n,n}(u), \quad a_n = A_n(1).$$

The empty bridge is counted once, including in dimension zero. Random bridges below are uniform over this finite set; they are equivalently uniform-step random walks conditioned to return to the origin. We write J_n for their zero-step count and K_n for the number of axes used at least once. An axis is counted once regardless of how many steps use it.

OEIS A328716 [1] records the exact diagonal formula, attributed to Ilya Gutkovskiy on 26 October 2019, and a leading equivalent with numerical parity constants, attributed to Václav Kotěšovec on 27 October 2019. The general array A328718 [2] identifies A328716 as its diagonal; its entry is due to Seiichi Manyama, and its walk interpretation is attributed to Alois P. Heinz. The classical Bessel-product walk generating function is given by Gessel, Weinstein, and Wilf [3], equation (2). Our lazy step appends an elementary exponential factor.

The large-powers method, including full descending-power expansions and compact-ratio uniformity, is classical; see Flajolet and Sedgewick [4], Theorem VIII.8 and its proof. Their subsequent remarks also discuss vanishing multipliers. We give complete arguments specialized to this period-two family, rather than interpreting the existence of a full expansion as a new analytic theorem. The explicit coefficients, marked probability refinements, occupation calculation, and inverse brackets are the objects of this report. No exhaustive historical novelty claim is made.

Every all-orders statement means a Poincaré expansion: the order M is fixed before the size parameter tends to infinity. Constants and onsets may depend on M and any specified compact sets. We do not assert convergence of the infinite series, allow M to grow, or supply computable remainder constants.

Remark 1.1 ([write] The OEIS entries, quoted, 7 October 2026). The entries were read on 7 October 2026 in the OEIS internal format; the quotations are verbatim.

- (a) A328716 (revision #38, 27 October 2019), offset 0, by Seiichi Manyama (26 October 2019):
“Constant term in the expansion of $(1 + x_1 + x_2 + \dots + x_n + 1/x_1 + 1/x_2 + \dots + 1/x_n)^n$.”; a comment by Alois P. Heinz (26 October 2019) giving the walk interpretation; the formulas “ $a(n) = n! * [x^n] \exp(x) * \text{BesselI}(0, 2*x)^n$. - `_Ilya Gutkovskiy_, Oct 26 2019`” ((2) with $N = D = n, u = 1$) and “ $a(n) \sim c * d^n * n^n$, where $d = 0.8047104059195202206625458331930618795\dots$ and $c = 2.12946224998808159475495497\dots$ if n is even and $c = 1.4189559976544232606562785\dots$ if n is odd. - `_Vaclav Kotesovec_, Oct 27 2019`”; a b-file “Table of $n, a(n)$ for $n = 0..398$ ” (Manyama; terms 0–199 from Heinz).
- (b) A328718 (revision #35, 30 October 2019), the array $T(n, k)$ with n the dimension and k the number of steps, by Manyama, with Heinz’s walk comment and the comment “**Conjecture:** Row r is asymptotic to $(2*r+1)^{(n+r/2)} / (2^r * (\pi*n)^{(r/2)})$. - `_Vaclav Kotesovec_, Oct 27 2019`” (proved in Remark 9.2), and column polynomials $T(n, 2), \dots, T(n, 7)$ (Kotěšovec, 30 October 2019).

Kotěšovec’s d and both values of c agree to every printed digit with d, C_+ and C_- of (7), computed by the write at 60 digits (his decimals are truncations); Theorem 4.1 at $M = 0$ proves his equivalent, for which the entry gives no proof. The write compared all 399 b-file terms with the exact integer

convolution of Section 10, and 44 of them ($n \leq 39$ and $n = 100, 199, 250, 398$) with a direct rational power of F , a different route: all agree, and a_n is strictly increasing from $n = 1$ with $a_0 = a_1$. The source's twenty displayed terms agree. Nothing was submitted to the OEIS.

1.1 Provenance, sources and reading conventions

[write] **Added 7 October 2026, batch 109.** *Provenance.* This report is Report 179 of the session bundle that ProveIt's docs/incoming intake processed as batch 109: the archive `Lazy_Closed_Walks_Asymptotics_and_Inverses_Source.zip` (617,713 bytes, SHA-256 `e433f40b68ba...e5473b44f613`, 26 files, no wrapper directory), arrival commit `60f54ea06`, placed in `f7e9e5c2f`. The text is the delivered `Report179.tex` (701 lines, dated 3 October 2026), shipped as `article.tex`. The core programs are shipped under `code/` (the root `build.py`, `guard_tests.py`, `test_build.py` and `verify_manifest.py` too), the optional ones as `code/optional-*.py`, the data and generated/ records under `data/`, `optional/README.md` as `optional-README.md`, and `README_CODE.md` and `SOURCE_AUDIT.md` at the root (the report README maps every name). Not shipped, and retrievable from the arrival commit: the delivered 16-page `Report179.pdf`, the checksum manifest `SHA256SUMS.json` (25 entries, verified at placement), and the delivered README, replaced by the report README. The package pins the ProveIt commit `83befe707` (3 October 2026), which exists; it carries no “prepared for private review” line and no personal data. It is a single-source report, so no merge choice was made. The write prefixed the 72 delivered labels with `lcw:` and updated the 65 references to them; it added this subsection, the notes marked [write] and Remarks 1.1, 8.2 and 9.2, each the last statement of its section, and inserted nothing before a delivered display, table or statement, so every theorem, section and equation of the source keeps its delivered number. The verified-prefix markers of Section 2, which `code/verify.py` reads, are unchanged.

The sources, as the write read them (7 October 2026).

- OEIS A328716 and A328718 in the internal format, and the A328716 b-file (Remark 1.1); the source read the official GitHub record of A328716 because the webpage did not load for it.
- The transseries volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`: `p0:def:three-inverses`, `p0:thm:staircase` (with `p0:rem:parity-instances`), `p0:prop:factorial-core`, `p0:thm:core-reversion`, `plt:def:lw-monomial-log-datum` and `plt:thm:lw-template`, read for Remark 8.2.
- Not read by the write: Gessel–Weinstein–Wilf [3] and Flajolet–Sedgewick [4]; the source's account of them (equation (2); Theorem VIII.8 and Note VIII.37) is reported, not confirmed.

What was checked at intake. The batch-109 dossier read the manuscript and found no error; with its own exact polynomial power at $n = 120, 121$ it found r , d , $C_{\pm}$ as printed and the scaled residuals consistent with $c_{\epsilon,1}$, $c_{\epsilon,2}$; it confirmed the pin and the audit's statements about the repository; and it reran the programs on copies: `code/verify.py` passing (its summary equal to `generated/verification.json` as JSON), `regenerate.py--compare` and both optional `--compare` checks exiting 0, `guard_tests.py` failing on Windows (text regenerated with CRLF). *What the write checked.* Every proof line by line, with no mathematical error found: the counting (2)–(3) and the injection; the Riccati equation (6) and $b = 4r^2 - 1$; $\kappa_3, \dots, \kappa_6$ from (8); the two-arc reduction and the parity of the odd coefficients; $\mathcal{T}_1$ and $c_{\epsilon,1}$ (21); the quotient and Cauchy steps of Theorem 6.1 and $Q_{\epsilon,1}$; Ψ 's derivatives, τ^2 and the mean offset (33); the parity brackets and the width argument of Theorem 8.1; the closed form (42) and the displacement; and the compactness argument of Theorem 9.1. With its own code (Python 3.14.4, mpmath 1.3.0): the constants of Sections 3 and 7 at 60 digits (note at the end of Section 3); $c_{\epsilon,1}$, $c_{\epsilon,2}$, $c_{\epsilon,3}$ from (21)–(23) against the table of Section 5; the diagnostics table of Section 10 to every printed digit; $\mathbb{E}K_n - np$ from exact marked counts; the b-file (Remark 1.1); and the row conjecture numerically (Remark 9.2). It reran the programs from the shipped files (README, Route B): `code/verify.py` passing with its summary equal to the shipped record as JSON, `regenerate.py--compare` and both optional checks exiting 0, `guard_tests.py` failing on Windows as at intake. Finite and floating computations are observations, not certificates.

Relation to the repository. Before batch 109 no file of the repository named A328716 or A328718. The source's audit, pinned at `83befe707`, read the A039831 report (now `a039831-two-fourier-peaks`, a moving maximum coefficient) and set aside A047909 and A108242 as covered (now `a047909-beta-renewal-`

subsequences and `a108242-regular-cyclic-word-covers`) and Report 169’s packed matrices (now Part IV of `a261781-matrix-compositions`), all in `oeis-sequence-asymptotics/`; none shares a result with this report, so no reciprocal note is proposed. No statement of this report is formalized in Lean or Rocq, and its place in the collection confers no formal status. The inverses of Section 8 are compared with the transseries volume in Remark 8.2.

[write] **What is not claimed, collected from the source.** The exact formula and the numerical leading equivalent are already in the OEIS; the Bessel walk generating function and the large-powers method, including full expansions, compact-ratio uniformity and vanishing multipliers, are classical; no open-problem resolution, no historical priority (“bounded evidence, not an absence certificate”), and full priority for the individual probability refinements “remains unestablished”. Every expansion is a Poincaré expansion at fixed order: no convergence, no growing order, no computable remainder constants or onsets, no certified finite-input inverse calculator (“The existential width-one theorem is not by itself a certified finite-input inverse calculator”). The weighted expansion is a signed approximation; the occupation limits are weak limits, not total-variation convergence to a Gaussian. Nothing treats $\rho \downarrow 0$, $\rho \rightarrow \infty$ or fixed D , and A328718’s row statement is a separate regime. The diagnostics are not interval-certified; no OEIS record or external repository was edited. The write adds: its own checks are floating or finite; Remark 9.2 proves a fixed-dimension statement by the classical local limit argument, with no novelty claimed; Remark 8.2 claims no novelty for any inversion.

Reading conventions (letters the source reuses, with the tempting false readings; no symbol of the source was renamed; symbols of the transseries volume carry the subscript “vol” in Remark 8.2).

Symbol	Meaning here (where)	Not to be read as
$H(v, t), H(s), H_\ell(j)$	the phase exponent (18); $h(e^s)$ (Section 7); the monomial polynomials (28)	each other
$h, h(w)$	a cumulant order or summation index; the moving log partition function (34)	each other; the volume’s h_{vol}
$R, R', R_2(n), R_{k,m}$	disk radii (Theorems 4.1, 6.1); the scaled residual (Section 10); occupied-axis counts	each other
$q, Q_{D,k}, Q_{\epsilon,\ell}$	$1/f$ (Section 7), also the even index in (10); integer convolutions (45); probability polynomials	each other
K_n, K	occupied axes; a compact ratio interval (Section 9)	each other
$D, \mathcal{D}$	the dimension; $z d/dz$ (1)	each other
$\rho, r, r_\rho, r(w)$	N/D ; the saddle; the ratio saddle; the moving saddle	the volume’s ρ_{vol} (a residue class or a datum)
$\mathcal{T}_\ell, T(n, k)$	saddle functionals (11); the A328718 array (n the dimension)	each other
d, d_{TV}	$F(r)/(er)$; total variation	each other; the volume’s d_{vol}
$E_\epsilon, E_{\epsilon,u}, \mathbb{E}$	$e^t + \epsilon e^{-t}$; the marked amplitude; expectation	each other
v_ϵ, v	$rE'_\epsilon(r)/E_\epsilon(r)$ (also v in (21)–(23)); $n^{-1/2}$ in (18)	each other
s_j, s	Stirling coefficients (12); a real variable (Section 7)	each other
$\ell, \ell_{\epsilon,j}$	an expansion order; logarithmic coefficients (37)	each other
$A, A_n(u), A_M$	an amplitude; the counts; the inverse constant (39)	the volume’s a_{vol}
$F, \mathcal{F}_{\epsilon,M}$	$I_0(2z)$; the inverse carriers (38)	each other
$X, X_r, x, x(s)$	the inverse argument; the tilted variable (4); z^2 ; $\log r(e^s)$	each other; the volume’s X_{vol}
I, I_0, I_1	the indicator $\mathbf{1}_{\{X>0\}}$; Bessel functions	each other
κ_h, Λ, L_M	cumulants; $1 + \log(dt)$; the lower bracket	the volume’s $\kappa_{\text{vol}}, \Lambda_{\text{vol}}, L_{\text{vol}}$

2 Exact enumeration and tilted coordinates

Put

$$F(z) = I_0(2z) = \sum_{k \geq 0} \frac{z^{2k}}{(k!)^2}, \quad \mathcal{D} = z \frac{d}{dz}. \quad (1)$$

The defining series is sufficient for all arguments; no Bessel asymptotic formula is required.

Proposition 2.1. *For nonnegative integers N, D ,*

$$A_{N,D}(u) = N![z^N]e^{uz}F(z)^D, \quad (2)$$

$$A_n(u, w) := \sum_{\text{bridges}} u^{J_n} w^{K_n} = n![z^n]e^{uz}\{1 + w(F(z) - 1)\}^n. \quad (3)$$

In particular $J_n \equiv n \pmod{2}$, $0 \leq K_n \leq \lfloor (n - J_n)/2 \rfloor$, and $a_{n+1} > a_n$ for every $n \geq 1$.

Proof. If there are j zero steps and k_i positive and k_i negative steps in axis i , then $j + 2 \sum_i k_i = N$ and the number of orderings is

$$\frac{N!}{j! \prod_i (k_i!)^2}.$$

Summing proves (2); marking each nonzero k_i proves (3). The parity and support assertions follow from the same constraint. To prove strict monotonicity, embed a bridge in $\mathbb{Z}^{n+1}$ and append a zero step. This is injective. A bridge using $+e_{n+1}$ once, $-e_{n+1}$ once, and $n - 1$ zero steps lies outside the image. The only initial equality is $a_0 = a_1 = 1$. $\square$

The first twenty values, beginning at $n = 0$, agree with the inspected OEIS record:

$$\begin{aligned} &1, 1, 5, 19, 217, 1451, 26041, 249705, 6116209, 76432627, \\ &2373097921, 36562658573, 1374991573825, 25188442156333, \\ &1112491608614933, 23620069750701091, 1198207214200181217, \\ &28930659427538020915, 1657461085278025906081, \\ &44848606508761385855085. \end{aligned}$$

For any $r > 0$ define an even-valued variable X_r by

$$\mathbb{P}(X_r = 2k) = \frac{r^{2k}}{(k!)^2 F(r)} \quad (k \geq 0). \quad (4)$$

Its cumulant generating function is $\log F(re^s) - \log F(r)$, so

$$\mathbb{E}X_r = \mathcal{D} \log F(r), \quad \text{Var } X_r = \mathcal{D}^2 \log F(r) > 0.$$

If $X_1, \dots, X_D$ have this law independently and $Z \sim \text{Poisson}(r)$ is independent, then, conditional on $Z + \sum_i X_i = N$, the pair $(Z, \sum_i \mathbf{1}_{\{X_i > 0\}})$ has exactly the zero-step and occupied-axis law of the uniform bridge. Indeed every admissible multiplicity vector has conditional mass proportional to $1/(j! \prod_i (k_i!)^2)$; its factor r^N cancels. This interpretation will explain the occupation variance.

3 The saddle and exact constants

Lemma 3.1. *There is a unique $r > 0$ such that*

$$\mathcal{D} \log F(r) = \frac{2r I_1(2r)}{I_0(2r)} = 1. \quad (5)$$

Writing $b = \mathcal{D}^2 \log F(r)$, one has $b = 4r^2 - 1 > 0$.

Proof. The mean in (4) is strictly increasing in $\log r$, since its derivative is the positive variance. It tends to zero as $r \downarrow 0$. It tends to infinity as $r \rightarrow \infty$: for any finite support cutoff, comparison with one higher positive term of F shows that the probability below the cutoff tends to zero. Thus the intermediate value theorem gives the unique solution.

Termwise differentiation of (1) gives $z^2 F'' + z F' - 4z^2 F = 0$. With $a(z) = \mathcal{D} \log F(z)$ this becomes

$$\mathcal{D}a = 4z^2 - a^2. \quad (6)$$

At $a(r) = 1$ it gives the asserted expression for b . $\square$

For $\epsilon \in \{+1, -1\}$ define

$$E_\epsilon(t) = e^t + \epsilon e^{-t}, \quad d = \frac{F(r)}{er}, \quad C_\epsilon = \frac{E_\epsilon(r)}{\sqrt{b}}, \quad v_\epsilon = \frac{r E'_\epsilon(r)}{E_\epsilon(r)}. \quad (7)$$

Both C_ϵ are positive. The signs $+1, -1$ always denote even and odd n , respectively. Thus $v_+ = r \tanh r$ and $v_- = r \coth r$. Useful numerical values are

$$\begin{aligned} r &= 0.804139735863439633472418655\dots, \\ b &= 1.586562859178089847374174939\dots, \\ d &= 0.804710405919520220662545833\dots, \\ C_+ &= 2.129462249988081594754954973\dots, \\ C_- &= 1.418955997654423260656278534\dots \end{aligned}$$

These reproduce the numerical leading constants already recorded in A328716. The decimal values are ordinary high-precision evaluations, not interval certificates.

All higher saddle cumulants have finite algebraic definitions. Set $x = z^2$, $a = \mathcal{D} \log F(z)$, and

$$\mathcal{L} = 2x\partial_x + (4x - a^2)\partial_a, \quad \kappa_h = \mathcal{L}^{h-1}a \Big|_{a=1, x=(b+1)/4}. \quad (8)$$

Then $\kappa_h = \mathcal{D}^h \log F(r)$ and, explicitly,

$$\begin{aligned} \kappa_1 &= 1, & \kappa_2 &= b, & \kappa_3 &= 2, \\ \kappa_4 &= -2b(b-2), & \kappa_5 &= 4(b-2)(b-1), \\ \kappa_6 &= 8(2b-3)(b^2-b+1). \end{aligned} \quad (9)$$

Equation (6) proves (8) inductively. In particular all cumulants are polynomials in b .

[write] Rechecked, 7 October 2026. Recomputed at 60 digits from (5): d , C_+ , C_- above and τ^2 in Section 7 are correct truncations, but the last printed digits of r , b and p are rounded up: the truncations are $r = 0.804139735863439633472418654\dots$ (next digits 99884), $b = 1.586562859178089847374174938\dots$ (92253) and, in Section 7, $p = 0.431494883273705551403382959\dots$ (59299). The delivered digits are kept; every printed digit except these three is a truncation.

4 An absolute expansion to every fixed order

For an amplitude A analytic near r , define a linear functional $\mathcal{T}_\ell A$ as follows. Sum over $j \geq 0$ and finitely supported nonnegative integers $(m_h)_{h \geq 3}$ satisfying

$$j + \sum_{h \geq 3} (h-2)m_h = 2\ell, \quad q = j + \sum_{h \geq 3} h m_h. \quad (10)$$

The integer $q = 2\ell + 2 \sum_h m_h$ is even. Set

$$\mathcal{T}_\ell A = \sum_{(10)} \frac{(\mathcal{D}^j A)(r)}{j!} \frac{(-1)^{q/2} (q-1)!!}{b^{q/2}} \prod_{h \geq 3} \frac{\kappa_h^{m_h}}{m_h! (h!)^{m_h}}. \quad (11)$$

Here $(-1)!! = 1$. The sum is finite, uses amplitude derivatives only through 2ℓ , and cumulants only through $2\ell + 2$. In particular $\mathcal{T}_0 A = A(r)$.

Let s_j be the classical Stirling coefficients, specified to all fixed orders by the formal identity

$$\sum_{j \geq 0} s_j t^j = \exp \left(\sum_{k \geq 1} \frac{B_{2k}}{2k(2k-1)} t^{2k-1} \right), \quad \frac{x}{e^x - 1} = \sum_{j \geq 0} B_j \frac{x^j}{j!}. \quad (12)$$

Thus $s_0 = 1$, $s_1 = 1/12$, $s_2 = 1/288$, $s_3 = -139/51840$. Define

$$\mathcal{P}_\ell = \sum_{h=0}^{\ell} s_h \mathcal{T}_{\ell-h}, \quad E_{\epsilon,u}(z) = e^{uz} + \epsilon e^{-uz}. \quad (13)$$

All series used for coefficient extraction in this report are formal when an infinite expansion is displayed.

Theorem 4.1. *For each fixed $M \geq 0$ and $R < \infty$, uniformly for complex $|u| \leq R$ as $n \rightarrow \infty$,*

$$A_n(u) = \frac{(dn)^n}{\sqrt{b}} \left\{ \sum_{\ell=0}^M \frac{\mathcal{P}_\ell E_{\epsilon,u}}{n^\ell} + O_{M,R}(n^{-M-1}) \right\}, \quad \epsilon = (-1)^n. \quad (14)$$

The error inside braces is absolute, including at zeros of $E_\epsilon(ru)$. In particular, with $c_{\epsilon,\ell} = \mathcal{P}_\ell E_{\epsilon,1}/E_\epsilon(r)$,

$$a_n = C_\epsilon (dn)^n \left\{ \sum_{\ell=0}^M \frac{c_{\epsilon,\ell}}{n^\ell} + O_M(n^{-M-1}) \right\}, \quad c_{\epsilon,0} = 1. \quad (15)$$

4.1 The two dominant arcs

Cauchy's formula on $|z| = r$ gives

$$[z^n] e^{uz} F(z)^n = \frac{F(r)^n r^{-n}}{2\pi} \int_{-\pi}^{\pi} e^{ure^{i\theta}} \left(\frac{F(re^{i\theta})}{F(r)} \right)^n e^{-in\theta} d\theta. \quad (16)$$

Equality in $|F(re^{i\theta})| \leq F(r)$ requires $e^{2ik\theta} = 1$ for every $k \geq 0$, since every even coefficient is strictly positive and the constant coefficient is nonzero. Hence it occurs only at $\theta = 0, \pi$ modulo 2π . Outside fixed small neighborhoods of these points the ratio is bounded by a constant $\gamma < 1$. The amplitude is uniformly bounded for $|u| \leq R$, so those arcs are exponentially small.

Near zero an analytic logarithm exists and

$$\phi(\theta) := \log \frac{F(re^{i\theta})}{F(r)} - i\theta = -\frac{b\theta^2}{2} + \sum_{h \geq 3} \frac{\kappa_h(i\theta)^h}{h!}. \quad (17)$$

By shrinking the neighborhood, $\Re \phi(\theta) \leq -b\theta^2/4$. Near π , translating the angle by π gives the identical phase and amplitude $\epsilon e^{-ure^{i\theta}}$. Thus the two local integrals combine into a single symmetric integral with amplitude $E_{\epsilon,u}(re^{i\theta})$.

4.2 A controlled integrated remainder

Fix any $0 < \delta < 1/2$ and restrict further to $|\theta| \leq n^{-1/2+\delta}$. The removed intermediate arcs contribute $O(\exp(-cn^{2\delta}))$ relative to the saddle scale, even after multiplication by any fixed power of n . Put $t = \sqrt{n}\theta$ and $v = n^{-1/2}$. After extracting $e^{-bt^2/2}$, the remaining phase exponent is

$$H(v, t) = \sum_{h \geq 3} \frac{\kappa_h(it)^h v^{h-2}}{h!}, \quad (18)$$

an analytic function of v at zero. On $|t| \leq n^\delta$, uniformly for $0 \leq v' \leq v$, convergence of the phase series yields

$$|H(v', t)| \leq Cv|t|^3 \leq Cn^{-1/2+\delta}t^2.$$

For all sufficiently large n the last constant is smaller than, for example, $b/8$. Each fixed derivative of H in v' is bounded by a fixed polynomial in $|t|$, uniformly on this set. This follows by differentiating its convergent series with $|v't|$ in a fixed sufficiently small disk. Likewise the k th v' derivative of $A(re^{iv't})$ is bounded by $C_k|t|^k$ for amplitudes in any compact analytic family.

The product $A(re^{iv't})e^{H(v',t)}$ therefore has every fixed v' derivative bounded by a polynomial in $|t|$ times $e^{bt^2/8}$. Taylor's formula through degree $2M+1$ in v has, after multiplication by $e^{-bt^2/2}$ and integration, a remainder

$$O(v^{2M+2}) \int_{\mathbb{R}} (1+|t|)^L e^{-3bt^2/8} dt = O(n^{-M-1}).$$

This bound is uniform for $A = E_{\epsilon,u}$, $|u| \leq R$. A coefficient of odd degree in v is an odd polynomial in t : in the notation (10), its total t degree differs from its v degree by an even integer. Such coefficients integrate to zero on the symmetric interval. Replacing each remaining finite-range Gaussian moment by the full moment has an exponentially small error.

The normalized Gaussian moments are

$$\frac{\int_{\mathbb{R}} (it)^q e^{-bt^2/2} dt}{\int_{\mathbb{R}} e^{-bt^2/2} dt} = (-1)^{q/2} (q-1)!! b^{-q/2} \quad (q \text{ even}).$$

Expanding the amplitude and the exponential in (18) now gives precisely (11), with coefficient constraint (10). We have proved

$$[z^n] e^{uz} F(z)^n = \frac{F(r)^n r^{-n}}{\sqrt{2\pi n b}} \left\{ \sum_{\ell=0}^M \frac{\mathcal{T}_\ell E_{\epsilon,u}}{n^\ell} + O_{M,R}(n^{-M-1}) \right\}. \quad (19)$$

Finally use the fixed-order Stirling expansion $n! = \sqrt{2\pi n} (n/e)^n (\sum_{j=0}^M s_j n^{-j} + O_M(n^{-M-1}))$. Multiplying gives (14). Since $E_\epsilon(r) > 0$, dividing only at $u=1$ proves (15). This completes the proof of Theorem 4.1.

Remark 4.2. For even n , $u = i\pi/(2r)$ makes $E_+(ru) = 0$; for odd n , $u = i\pi/r$ makes $E_-(ru) = 0$. Formula (14) still controls the first nonzero correction there. A relative error of the form $E_\epsilon(ru)(1 + O(n^{-1}))$ would not give this statement and generally would be false at these points. No division by a marked leading amplitude occurs in the proof.

5 The first three explicit corrections

The normalized amplitude derivatives are the moments of the limiting parity law that will appear in Section 6. Direct differentiation gives

$$\mu_j := \frac{(\mathcal{D}^j E_{\epsilon,1})(r)}{E_\epsilon(r)} = \sum_{h=0}^j \left\{ \begin{matrix} j \\ h \end{matrix} \right\} \left(\frac{b+1}{4} \right)^{[h/2]} \begin{cases} 1, & h \text{ even,} \\ v_\epsilon, & h \text{ odd,} \end{cases} \quad (20)$$

with $\mu_0 = 1$ and Stirling numbers of the second kind in braces. This follows from the Touchard identity $(z\partial_z)^j e^z = e^z \sum_h \left\{ \begin{matrix} j \\ h \end{matrix} \right\} z^h$ and $r^2 = (b+1)/4$.

Equations (8), (11), (12), and (20) constitute a finite exact algorithm at arbitrary fixed order. In the following formulas put $v = v_\epsilon$:

$$c_{\epsilon,1} = \frac{9-7b}{24b} - \frac{5}{6b^3} + \frac{(2-b)v}{2b^2}. \quad (21)$$

For readability write the next two as polynomials divided by their common denominator:

$$\begin{aligned} 1152b^6 c_{\epsilon,2} &= 193b^6 + 312b^5v - 594b^5 - 2856b^4v \\ &\quad + 1245b^4 + 4368b^3v + 1624b^3 + 3360b^2v \\ &\quad - 6552b^2 - 6720bv + 6160, \end{aligned} \quad (22)$$

$$\begin{aligned} 414720b^9 c_{\epsilon,3} &= -37427b^9 + 105660b^8v + 222507b^8 \\ &\quad + 934560b^7v - 1371537b^7 - 8147700b^6v \\ &\quad + 2306661b^6 + 8541000b^5v + 8767656b^5 \\ &\quad + 23476320b^4v - 22863852b^4 - 36146880b^3v \\ &\quad - 9517200b^3 - 14414400b^2v + 45405360b^2 \\ &\quad + 28828800bv - 27227200. \end{aligned} \quad (23)$$

They evaluate as follows:

Parity	$c_{\epsilon,1}$	$c_{\epsilon,2}$	$c_{\epsilon,3}$
Even	-0.219966033027	0.070696999974	-0.011375589727
Odd	-0.164864931187	0.080945110173	-0.071114538915

For example, the first saddle functional is

$$\mathcal{T}_1 A = -\frac{\mathcal{D}^2 A(r)}{2b} + \frac{\kappa_3 \mathcal{D} A(r)}{2b^2} + \left(\frac{\kappa_4}{8b^2} - \frac{5\kappa_3^2}{24b^3} \right) A(r). \quad (24)$$

Adding the Stirling term $A(r)/12$, substituting (9), and using $\mu_1 = v$, $\mu_2 = r^2 + v$ yields (21). The finite algorithm gives (22)–(23); an independent formal-exponential recurrence provides an algebraic cross-check described in Section 10.

6 Zero steps and weighted probability expansions

Let J_ϵ be $\text{Poisson}(r)$ conditioned on $(-1)^{J_\epsilon} = \epsilon$. Write its mass function and probability generating function as

$$\pi_\epsilon(j) = \begin{cases} \frac{2r^j}{j!E_\epsilon(r)}, & (-1)^j = \epsilon, \\ 0, & \text{otherwise,} \end{cases} \quad \mathbb{E}u^{J_\epsilon} = \frac{E_\epsilon(ru)}{E_\epsilon(r)}. \quad (25)$$

In particular $\mathbb{E}J_\epsilon = v_\epsilon$ and $\mathbb{E}J_\epsilon^2 = r^2 + v_\epsilon$.

Theorem 6.1. *For every fixed $M \geq 0$ and $R > 1$, there are polynomials $Q_{\epsilon,\ell}(j)$ of degree at most 2ℓ , with $Q_{\epsilon,0} = 1$, such that*

$$\sum_{j \geq 0} R^j \left| \mathbb{P}(J_n = j) - \pi_\epsilon(j) \sum_{\ell=0}^M \frac{Q_{\epsilon,\ell}(j)}{n^\ell} \right| = O_{M,R}(n^{-M-1}), \quad \epsilon = (-1)^n. \quad (26)$$

For $\ell \geq 1$, $\mathbb{E}Q_{\epsilon,\ell}(J_\epsilon) = 0$. The first polynomial is

$$Q_{\epsilon,1}(j) = -\frac{j^2 - r^2 - v_\epsilon}{2b} + \frac{j - v_\epsilon}{b^2}. \quad (27)$$

Consequently the total-variation distance from J_ϵ is $O(n^{-1})$, and every fixed polynomial moment has an expansion to all fixed orders.

Proof. Choose $R' > R$. Theorem 4.1 holds uniformly on $|u| \leq R'$. Dividing by the expansion at $u = 1$ is legitimate since its leading coefficient is the positive $E_\epsilon(r)$. Finite series division gives a probability-generating-function expansion with an absolute remainder uniformly $O(n^{-M-1})$ on that disk.

To identify its coefficients explicitly, define the monomial polynomials

$$H_\ell(j) = r^{-j} \mathcal{P}_\ell(z^j). \quad (28)$$

Since $\mathcal{D}^k z^j = j^k z^j$, these are polynomials in j of degree at most 2ℓ , and

$$c_{\epsilon,\ell} = \mathbb{E}H_\ell(J_\epsilon).$$

The probability polynomials are specified by the finite recurrence

$$Q_{\epsilon,0} = 1, \quad Q_{\epsilon,\ell}(j) = H_\ell(j) - \sum_{h=1}^{\ell} c_{\epsilon,h} Q_{\epsilon,\ell-h}(j). \quad (29)$$

Indeed the coefficient of u^j in $\mathcal{P}_\ell E_{\epsilon,u}$ is $(1 + \epsilon(-1)^j)r^j H_\ell(j)/j!$. Dividing by the unmarked denominator gives (29), including exact zeros at the wrong parity. Taking expectation in that recurrence proves centering inductively.

If $\mathcal{R}_n(u)$ denotes the analytic remainder after quotient expansion, Cauchy's inequality gives

$$|[u^j]\mathcal{R}_n(u)| \leq Cn^{-M-1}(R')^{-j}.$$

Multiplication by R^j and summation of the geometric series proves (26). This step controls the infinite tail; pointwise coefficient convergence alone would not suffice.

Finally (24), $\kappa_3 = 2$, and the subtraction of the mean in (29) give (27). Total variation is one half the unweighted ℓ^1 norm. Any fixed polynomial is bounded in absolute value by a constant times R^j , giving the moment statement. $\square$

The finite correction in (26) is a signed approximation; it is not asserted to be a probability mass function term by term. Setting $M = 0$ shows $J_n \Rightarrow J_\epsilon$ along each parity subsequence. The whole sequence has no single zero-step weak limit, since even and odd support remain disjoint. In particular the zero steps stay of order one even though the walk length and dimension both grow.

7 Occupied axes and a Gaussian product limit

Put

$$f = F(r), \quad q = f^{-1}, \quad p = 1 - q, \quad \tau^2 = pq - \frac{q^2}{b}. \quad (30)$$

Numerically,

$$p = 0.431494883273705551403382960\dots, \quad \tau^2 = 0.041597460122151058557181923\dots$$

Theorem 7.1. *One has $\tau^2 > 0$ and*

$$\frac{K_n - np}{\sqrt{n}} \Rightarrow \mathcal{N}(0, \tau^2). \quad (31)$$

Along a fixed parity $\epsilon = (-1)^n$, jointly

$$\left(\frac{K_n - np}{\sqrt{n}}, J_n \right) \Rightarrow (G, J_\epsilon), \quad (32)$$

where $G \sim \mathcal{N}(0, \tau^2)$ is independent of J_ϵ . Moreover

$$\mathbb{E}K_n = np - \frac{q}{b} \left(v_\epsilon + \frac{b-1}{2} - \frac{1}{b} \right) + O(n^{-1}), \quad \text{Var } K_n = n\tau^2 + O(1). \quad (33)$$

The constant mean offsets are approximately -0.071243028912 for even n and -0.311664057977 for odd n .

7.1 An analytic moving saddle

Let

$$B_w(z) = 1 + w(F(z) - 1).$$

Near $w = 1$, the equation $\mathcal{D} \log B_w(r(w)) = 1$ has an analytic solution with $r(1) = r$. The implicit-function theorem applies in the variable $\log r$, because the derivative is $b > 0$. Shrink to a fixed complex neighborhood of $w = 1$ in which $r(w)$, $B_w(r(w))$, and

$$b(w) = \mathcal{D}^2 \log B_w(r(w))$$

are nonzero and $\Re b(w)$ is bounded below by a positive constant. Take logarithm and square-root branches from their real positive values at $w = 1$, and put

$$h(w) = \log B_w(r(w)) - \log r(w). \quad (34)$$

For completeness the moving contour is valid also for complex w . The parametrization $z = r(w)e^{i\theta}$ describes a positively oriented circle of radius $|r(w)|$, merely with a rotated starting point. The coefficient integrand is entire except for its pole at zero, so Cauchy's formula applies on this circle.

At $\theta = 0$ the exact saddle equation cancels the linear term and gives quadratic term $-b(w)\theta^2/2$. Since B_w is even, the second saddle is exactly $-r(w)$ and its phase factor is still ϵ .

At $w = 1$ the compact complement of the two local arcs has a strict normalized modulus gap. Continuity preserves it uniformly for all complex w in a sufficiently small closed neighborhood of one. Local logarithms exist on the two arcs; a global logarithm is not needed. The integrated Taylor-remainder argument of Section 4.2 then holds with complex analytic cumulants, uniformly in w and bounded u . The Gaussian integral is $\sqrt{2\pi/b(w)}$ with the specified branch, by analytic continuation in the right half-plane or direct Gaussian integration. In particular,

$$A_n(u, w) = \left(\frac{n}{e}\right)^n \frac{e^{nh(w)}}{\sqrt{b(w)}} \left\{ E_\epsilon(ur(w)) + O(n^{-1}) \right\}. \quad (35)$$

The remainder inside braces is absolute and analytic in u, w . The full fixed-order version follows in the same way. Cauchy's derivative estimate on a smaller w disk allows its uniform remainders to be differentiated any fixed number of times.

7.2 The variance density and limiting independence

Write $H(s) = h(e^s)$ and $x(s) = \log r(e^s)$. At $s = 0$ consider the pair (X, I) , where $X = X_r$ and $I = \mathbf{1}_{\{X > 0\}}$. The log partition function is

$$\Psi(s, x) = \log(1 + e^s(F(e^x) - 1)).$$

The saddle constraint is $\Psi_x(s, x(s)) = 1$. At $(0, \log r)$ its derivatives are

$$\Psi_s = p, \quad \Psi_{ss} = pq, \quad \Psi_{xx} = b, \quad \Psi_{sx} = \text{Cov}(I, X) = q.$$

The covariance identity follows from $IX = X$ and $\mathbb{E}X = 1$. Differentiating the constraint and $H(s) = \Psi(s, x(s)) - x(s)$ gives

$$x'(0) = -\frac{q}{b}, \quad H'(0) = p, \quad H''(0) = pq - \frac{q^2}{b} = \tau^2. \quad (36)$$

Strict Cauchy-Schwarz makes $\tau^2 > 0$, since I is not an affine function of X on its positive-probability support $0, 2, 4, \dots$

For real t, y , put $w = e^{it/\sqrt{n}}$ and $u = e^{iy}$ in (35). Divide by $A_n(1, 1)$ and multiply by $e^{-itp\sqrt{n}}$. Taylor expansion gives

$$n\{H(it/\sqrt{n}) - H(0)\} - itp\sqrt{n} = -\frac{\tau^2 t^2}{2} + O(n^{-1/2}).$$

The remaining amplitude tends to $E_\epsilon(re^{iy})/E_\epsilon(r)$, including at its zeros because the remainder was absolute. The joint characteristic functions therefore tend to

$$e^{-\tau^2 t^2/2} \frac{E_\epsilon(re^{iy})}{E_\epsilon(r)}.$$

This is the characteristic function of the independent product in (32); Lévy's continuity theorem applies. Setting $y = 0$ proves (31) without a parity restriction. These are weak limits, not total-variation convergence from a lattice variable to a continuous Gaussian.

7.3 The constant mean correction

At $u = 1$ the leading amplitude in (35) is nonzero in a small w neighborhood. Its analytic logarithm yields

$$\log \frac{A_n(1, e^s)}{A_n(1, 1)} = n\{H(s) - H(0)\} + \log \frac{E_\epsilon(r(e^s))}{E_\epsilon(r)} - \frac{1}{2} \log \frac{b(e^s)}{b} + O(n^{-1}).$$

The remainder may be differentiated, as established above. At fixed x , differentiating Ψ_{xx} with respect to s gives $\Psi_{xxs} = q(b-1)$. One direct check is $\mathcal{D}p = q$ and $\mathcal{D}^2p = q(b-1)$ at the saddle. Allowing x to move then gives, using $\kappa_3 = 2$,

$$\left. \frac{d}{ds} b(e^s) \right|_{s=0} = q(b-1) + \kappa_3 x'(0) = q \left(b-1 - \frac{2}{b} \right).$$

The derivative of $\log E_\epsilon(r(e^s))$ is $v_\epsilon x'(0)$. Together with (36) these prove the expectation formula in (33). Two derivatives prove the variance statement. The same analytic expansion supplies all fixed cumulant corrections, although only the first mean offset is needed here.

8 Parity sensitive inversion

Let

$$\nu(X) = \min\{n \geq 0 : a_n \geq X\}, \quad X \rightarrow \infty.$$

Strict monotonicity from Proposition 2.1 makes this a genuine threshold. For each parity let $\nu_\epsilon(X)$ be the first such integer of parity ϵ , so $\nu(X) = \min_\epsilon \nu_\epsilon(X)$.

Define the logarithmic coefficients formally by

$$\ell_{\epsilon,j} = [t^j] \log \left(\sum_{k \geq 0} c_{\epsilon,k} t^k \right). \quad (37)$$

For example $\ell_{\epsilon,1} = c_{\epsilon,1}$, $\ell_{\epsilon,2} = c_{\epsilon,2} - c_{\epsilon,1}^2/2$, and $\ell_{\epsilon,3} = c_{\epsilon,3} - c_{\epsilon,1}c_{\epsilon,2} + c_{\epsilon,1}^3/3$. For fixed $M \geq 0$ put

$$\mathcal{F}_{\epsilon,M}(t) = t \log(dt) + \log C_\epsilon + \sum_{j=1}^M \ell_{\epsilon,j} t^{-j}. \quad (38)$$

It is eventually increasing, with derivative $1 + \log(dt) + O_M(t^{-2})$, and tends to infinity. For large X let $t_{\epsilon,M}(X)$ be its unique sufficiently large solution of $\mathcal{F}_{\epsilon,M}(t) = \log X$. Write $\lceil t \rceil_\epsilon$ for the least integer at least t with $(-1)^n = \epsilon$.

Theorem 8.1. *For each fixed M there exists $A_M > 0$ such that, for all sufficiently large X , with*

$$\eta_\epsilon = \frac{A_M t_{\epsilon,M}^{-M-1}}{1 + \log(dt_{\epsilon,M})}, \quad (39)$$

one has

$$\lceil t_{\epsilon,M} - \eta_\epsilon \rceil_\epsilon \leq \nu_\epsilon(X) \leq \lceil t_{\epsilon,M} + \eta_\epsilon \rceil_\epsilon. \quad (40)$$

Taking the minimum of the two lower endpoints and of the two upper endpoints gives integers $L_M(X), U_M(X)$ satisfying

$$L_M(X) \leq \nu(X) \leq U_M(X), \quad U_M(X) - L_M(X) \in \{0, 1\} \quad (41)$$

eventually. No effective value of A_M or onset is asserted.

Proof. Taking logarithms in (15) gives, on each parity,

$$\log a_n = \mathcal{F}_{\epsilon, M}(n) + O_M(n^{-M-1}).$$

The leading relative factor is positive and tends to one, so the logarithm and its fixed-order remainder are valid. On a fixed-width neighborhood of $t = t_{\epsilon, M}$ the carrier derivative is comparable to $\Lambda = 1 + \log(dt)$, and n^{-M-1} is comparable to t^{-M-1} . Choose A_M strictly larger than the common logarithmic error bound after these comparisons.

The parity integer immediately preceding $\lceil t - \eta_\epsilon \rceil_\epsilon$ is below $t - \eta_\epsilon$ and within a fixed distance of t . By the mean value theorem its carrier is smaller than $\log X$ by more than the possible logarithmic error, so its actual count is below X . The integer $\lceil t + \eta_\epsilon \rceil_\epsilon$ has count at least X by the same argument with opposite sign. Monotonicity proves (40), including endpoint cases. Taking minima is valid because the global threshold is the minimum of the two parity thresholds.

It remains to prove the width assertion. At a common large argument the two carriers differ by $\log(C_+/C_-) + O_M(t^{-1})$. Their derivatives are of order $\log t$, so their roots differ by $O(1/\log t)$; their radii tend to zero. The convex hull of both real intervals $[t_{\epsilon, M} - \eta_\epsilon, t_{\epsilon, M} + \eta_\epsilon]$ therefore eventually has length less than one.

If $U_M > L_M$, choose a parity attaining the minimum lower ceiling L_M . Its upper ceiling must exceed L_M , so its real interval contains the integer L_M and extends above it. Every point of the other real interval is strictly below $L_M + 1$, since the common hull has length less than one. The opposite-parity upper ceiling is consequently at most $L_M + 1$. Thus $U_M \leq L_M + 1$. The alternative is $U_M = L_M$, completing (41). $\square$

For $M = 0$ the carriers have a closed form. With W_0 the positive real Lambert branch and $Y_\epsilon = \log(X/C_\epsilon)$,

$$t_{\epsilon, 0}(X) = \frac{Y_\epsilon}{W_0(dY_\epsilon)}. \quad (42)$$

Substitution verifies the equation exactly. If instead one starts from $t_* = (\log X)/W_0(d \log X)$, the first displacement is

$$t_{\epsilon, 0} - t_* = -\frac{\log C_\epsilon}{1 + \log(dt_*)} + o\left(\frac{1}{\log t_*}\right).$$

The two constants cannot be silently merged for sharp parity rounding. Newton iteration on (38) computes higher real carriers; interval rounding still requires (40). A ceiling is certified only when its two parity endpoints coincide, or the global endpoints coincide. The existential width-one theorem is not by itself a certified finite-input inverse calculator.

Remark 8.2 ([write] The inversions and the transseries volume, 7 October 2026). Compared, statement by statement, with the transseries volume (`Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`); symbols of the volume carry the subscript “vol”.

- (a) *Instance.* The threshold $\nu(X)$ is the integer staircase N_* of `p0: def: three-inverses` with $A_n = a_n$ and $n_1 = 1$ (strictly increasing by Proposition 2.1), for $X > 1$, and the decomposition $\nu = \min_\epsilon \nu_\epsilon$ is `p0: thm: staircase(4)` with $r_{\text{vol}} = 2$: the parity ceiling $\lceil t \rceil_\epsilon$ is $\rho_{\text{vol}} + 2 \lceil (t - \rho_{\text{vol}})/2 \rceil$ for the residue $\rho_{\text{vol}} \in \{0, 1\}$ with $(-1)^{\rho_{\text{vol}}} = \epsilon$, as in `p0: rem: parity-instances`, and each ν_ϵ is the parity ceiling of the inverse of any admissible interpolation of its class.
- (b) *Theorem 8.1:* the parity brackets (40) are an analogue as stated and proved; also a consequence of `p0: thm: staircase(4)` once interpolations are supplied. The source brackets each ν_ϵ directly about the root $t_{\epsilon, M}$ of the carrier, which is not the inverse of an interpolation

of the counts; inside each class the conclusion has the form of `p0:thm:staircase(2)`. A second route (added by the write): with $r_n = \log a_n - \mathcal{F}_{\epsilon,M}(n) = O(n^{-M-1})$ on the class, let $\tilde{r}_\epsilon$ interpolate (r_n) linearly between consecutive class members and $\mathcal{A}_\epsilon = \mathcal{F}_{\epsilon,M} + \tilde{r}_\epsilon$. Since $\mathcal{F}'_{\epsilon,M} \asymp \log t$ and $|\tilde{r}'_\epsilon| \leq \max |r_n|$, $\mathcal{A}_\epsilon$ is eventually continuous and strictly increasing with $\mathcal{A}_\epsilon(n) = \log a_n$ on the class, so it is admissible there; for its inverse ν^ϵ at $\log X$, $|\nu^\epsilon - t_{\epsilon,M}| \leq |\tilde{r}_\epsilon(\nu^\epsilon)| / \min \mathcal{F}'_{\epsilon,M} = O(t^{-M-1} / \log t)$, at most η_ϵ for a suitable A_M , and the monotonicity of $[\cdot]_\epsilon$ gives (40). The width statement (41) is the source's own; the volume asserts nothing of the kind.

- (c) *Instance.* The closed form (42) is `p0:prop:factorial-core` with $\kappa_{\text{vol}} = 1$, $d_{\text{vol}} = \log d$ and $L_{\text{vol}} = Y_\epsilon$: there $v = W_0(dY_\epsilon)$ and $X_{\text{vol}} = e^v/d = Y_\epsilon/v$, with the side condition $dt > 1$; the displacement $t_{\epsilon,0} - t_*$ is the first-order term given by the core's slope $1/(\kappa_{\text{vol}}(v+1)) = 1/(1 + \log(dt))$. Its large- Y expansion is a *formal* instance of `plt:thm:lw-template` with $Z_{\text{vol}} = \log t$, $\xi_{\text{vol}} = \log Y_\epsilon$ and data $(\rho_{\text{vol}}, \mu_{\text{vol}}, \beta_{\text{vol}}, B_{\text{vol}}) = (1, 1, 0, \log(1 + t_{\text{vol}} \log d))$, since $\log(t \log(dt)) = Z_{\text{vol}} + \log Z_{\text{vol}} + \log(1 + \log d/Z_{\text{vol}})$.
- (d) *Formal instance after a change of variable.* For $M \geq 1$, with $t = t_{\epsilon,0}(1 + E)$, $t_{\text{vol}} = 1/t_{\epsilon,0}$ and $L = \log(dt_{\epsilon,0})$, the carrier equation $\mathcal{F}_{\epsilon,M}(t) = \log X$ divided by $t_{\epsilon,0}L$ reads $\Lambda_{\text{vol}}E + h_{\text{vol}}(E) + f_{\text{vol}}(t_{\text{vol}}, E) = 0$ with $\Lambda_{\text{vol}} = 1 + 1/L$, $h_{\text{vol}}(u) = \{(1+u)\log(1+u) - u\}/L$ and $f_{\text{vol}} = \sum_{j=1}^M \ell_{\epsilon,j} t_{\text{vol}}^{j+1} (1+u)^{-j}/L$, which is `p0:thm:core-reversion` over the ring generated over $\mathbb{Q}$ by the $\ell_{\epsilon,j}$, L^{-1} and $(1+L)^{-1}$. The source computes the carriers by Newton iteration instead and states no such series.
- (e) *Not shown to be an instance.* For $M \geq 1$ the full carrier equation is not shown to be an instance of `plt:thm:lw-template`: in the chart of (c) the terms $\ell_{\epsilon,j} t^{-j}$ are of relative size $e^{-(j+1)Z_{\text{vol}}}$, not a series in $t_{\text{vol}} = 1/Z_{\text{vol}}$ with coefficients polynomial in $\log Z_{\text{vol}}$, so (H3) fails there. The write does not claim that no other chart makes it an instance.

No novelty is claimed for any of these inversions.

9 Proportional length and dimension

Let $D \rightarrow \infty$ and N be an integer such that $\rho = N/D$ belongs to a fixed compact interval $K \subset (0, \infty)$. Define $r_\rho > 0$ by

$$\mathcal{D} \log F(r_\rho) = \rho, \quad b_\rho = \mathcal{D}^2 \log F(r_\rho) > 0, \quad \kappa_{\rho,h} = \mathcal{D}^h \log F(r_\rho).$$

The argument of Lemma 3.1 proves existence and uniqueness. Compactness of K puts the saddles in a compact positive interval and bounds b_ρ away from zero. Let $\mathcal{T}_{\rho,\ell}$ denote (11) with r, b, κ_h replaced by these quantities.

Theorem 9.1. *For fixed $M \geq 0$, $R < \infty$, uniformly for $\rho \in K$ and complex $|u| \leq R$,*

$$A_{N,D}(u) = \frac{N! F(r_\rho)^D r_\rho^{-N}}{\sqrt{2\pi D b_\rho}} \left\{ \sum_{\ell=0}^M \frac{\mathcal{T}_{\rho,\ell} E_{\epsilon,u}}{D^\ell} + O_{M,R,K}(D^{-M-1}) \right\}, \quad \epsilon = (-1)^N. \quad (43)$$

The remainder is absolute inside braces. Here $E_{\epsilon,u}(z) = e^{uz} + \epsilon e^{-uz}$ as before; $N!$ is retained exactly.

Let $J_{N,D}$ count zero steps. It has an all-fixed-orders weighted ℓ^1 expansion, uniform for $\rho \in K$, about $\text{Poisson}(r_\rho)$ conditioned to have parity N . In particular, for every fixed $R > 1$,

$$\sum_{j \geq 0} R^j |\mathbb{P}(J_{N,D} = j) - \pi_{\rho,\epsilon}(j)| = O_{R,K}(D^{-1}), \quad (44)$$

where $\pi_{\rho,\epsilon}$ is (25) with r replaced by r_ρ . If additionally $\rho \rightarrow \rho_0 \in (0, \infty)$ and N has fixed parity ϵ , then $J_{N,D}$ converges weakly to $\text{Poisson}(r_{\rho_0})$ conditioned on that parity.

Proof. Use the circle $|z| = r_\rho$ and centered phase

$$\log \frac{F(r_\rho e^{i\theta})}{F(r_\rho)} - i\rho\theta.$$

The quadratic coefficient is $-b_\rho/2$, uniformly negative. Positivity and the even support give exactly two maximum-modulus points for every ρ ; compactness supplies a uniform strict gap away from the two fixed neighborhoods. Local analytic derivatives and amplitude bounds are uniform on compact sets. The argument of Section 4.2, scaled by $\sqrt{D}$, therefore proves (43). Translation through π contributes $e^{-iN\pi} = (-1)^N$, independent of the parity of D .

The unmarked leading amplitudes $E_\epsilon(r_\rho)$ are bounded away from zero for both signs on K . Quotient expansion and the larger-disk Cauchy argument of Theorem 6.1 prove (44) and all its fixed-order refinements. More explicitly set

$$H_{\rho,\ell}(j) = r_\rho^{-j} \mathcal{T}_{\rho,\ell}(z^j), \quad c_{\rho,\epsilon,\ell} = \mathbb{E} H_{\rho,\ell}(J_{\rho,\epsilon}),$$

where $J_{\rho,\epsilon}$ has mass $\pi_{\rho,\epsilon}$, and apply the recurrence (29) with these quantities. The resulting polynomials of degree at most 2ℓ give an error $O_{M,R,K}(D^{-M-1})$. Finally continuity of r_ρ gives the asserted fixed-law weak limit when ρ converges and parity is fixed. $\square$

If desired, expanding $N!$ in (43) replaces $\mathcal{T}_{\rho,\ell}$ by $\sum_{h=0}^{\ell} s_h \rho^{-h} \mathcal{T}_{\rho,\ell-h}$ after extracting the corresponding Stirling leading factor. Compactness controls the additional remainder. For a moving, nonconvergent ρ , (44) is an approximation by a moving parameter, not convergence to one fixed distribution. Nothing here treats $\rho \downarrow 0$, $\rho \rightarrow \infty$, or fixed D ; in particular the fixed-dimensional row statement in A328718 is a separate regime.

Remark 9.2 ([write] The rows of A328718, 7 October 2026). The entry A328718 states as a conjecture (Remark 1.1(b)) that its row r , the walks in fixed dimension $D = r$, satisfies

$$A_{N,D}(1) \sim \frac{(2D+1)^{N+D/2}}{2^D(\pi N)^{D/2}} \quad (N \rightarrow \infty, D \geq 0 \text{ fixed}).$$

This is true; it is the local central limit theorem for the lazy walk, and the short proof is as follows. For $D = 0$ both sides equal one. For $D \geq 1$, $A_{N,D}(1) = (2D+1)^N \mathbb{P}(S_N = 0)$ for the walk S_N with independent steps uniform on $\{0, \pm e_1, \dots, \pm e_D\}$, and by Fourier inversion

$$\mathbb{P}(S_N = 0) = (2\pi)^{-D} \int_{[-\pi,\pi]^D} \varphi(\theta)^N d\theta, \quad \varphi(\theta) = \frac{1 + 2 \sum_{i=1}^D \cos \theta_i}{2D+1}.$$

Here φ is real and $|\varphi| \leq 1$; $\varphi(\theta) = 1$ only at $\theta = 0$, and $\varphi = -1$ never, because $1 + 2 \sum_i \cos \theta_i \geq 1 - 2D > -(2D+1)$. So $|\varphi| \leq \gamma < 1$ outside a ball $|\theta| \leq \delta$, which contributes $O(\gamma^N)$. Inside it, $\varphi(\theta) = 1 - |\theta|^2/(2D+1) + O(|\theta|^4)$, so for small δ one has $0 < \varphi(\theta) \leq \exp(-|\theta|^2/(2(2D+1)))$, and after $\theta = x/\sqrt{N}$ dominated convergence gives $N^{D/2} \int_{|\theta| \leq \delta} \varphi^N d\theta \rightarrow \int_{\mathbb{R}^D} e^{-|x|^2/(2D+1)} dx = (\pi(2D+1))^{D/2}$. Hence $\mathbb{P}(S_N = 0) \sim (2\pi)^{-D} (\pi(2D+1)/N)^{D/2} = (2D+1)^{D/2} / (2^D(\pi N)^{D/2})$, which is the displayed formula. Numerically, from exact counts, $N(A_{N,D}(1) 2^D(\pi N)^{D/2} (2D+1)^{-N-D/2} - 1) = -0.1875, -0.3747, -0.5614$ at $N = 400$ for $D = 1, 2, 3$, consistent with a $1/N$ correction. Row 1 is A002426, the central trinomial coefficients. The argument is classical and is a fixed-dimension statement: it neither uses nor extends Theorem 9.1, whose compact-ratio regime excludes fixed D .

10 Exact checks and reproducibility

The proofs above establish the asymptotic results. Computation checks finite enumeration and coefficient algebra; numerical agreement does not establish a uniform remainder, an effective onset, or priority over prior literature.

10.1 Three finite counting descriptions

An integer convolution avoids cancellation. Put

$$Q_{D,k} = (k!)^2 [t^k] \left(\sum_{h \geq 0} \frac{t^h}{(h!)^2} \right)^D. \quad (45)$$

Then $Q_{0,0} = 1$, $Q_{0,k} = 0$ for $k > 0$, and

$$Q_{D,k} = \sum_{h=0}^k \binom{k}{h}^2 Q_{D-1,k-h}. \quad (46)$$

For $N - j = 2k \geq 0$, the exact number of bridges with j zero steps is

$$\binom{N}{j} \binom{2k}{k} Q_{D,k}. \quad (47)$$

The count is zero at other parities. To count occupied axes independently, put

$$R_{k,m} = (m!)^2 [t^m] \left(\sum_{h \geq 1} \frac{t^h}{(h!)^2} \right)^k.$$

These satisfy $R_{0,0} = 1$ and $R_{k,m} = \sum_{h=1}^m \binom{m}{h}^2 R_{k-1,m-h}$. For $n - j = 2m$,

$$\#\{J_n = j, K_n = k\} = \binom{n}{j} \binom{2m}{m} \binom{n}{k} R_{k,m}. \quad (48)$$

Both recurrences follow by selecting the contribution of one factor in the corresponding generating function.

A different exact calculation uses (6). Write $a(z) = \sum_{k \geq 1} \alpha_k z^{2k}$ and $F(z)^D = \sum_{k \geq 0} \beta_{D,k} z^{2k}$. Then

$$\alpha_k = \frac{4\mathbf{1}_{\{k=1\}} - \sum_{h=1}^{k-1} \alpha_h \alpha_{k-h}}{2k}, \quad (49)$$

$$\beta_{D,0} = 1, \quad \beta_{D,k} = \frac{D}{2k} \sum_{h=1}^k \alpha_h \beta_{D,k-h}. \quad (50)$$

The unmarked count is $N! \sum_{k=0}^{\lfloor N/2 \rfloor} \beta_{D,k} / (N - 2k)!$. Rational arithmetic makes this comparison exact.

10.2 Independent coefficient verification

A raw-derivative route to the cumulants differentiates the second-order Bessel equation instead of iterating (8). Let $g_m = r^m F^{(m)}(r)/F(r)$, with $g_0 = g_1 = 1$. Then

$$g_{m+2} = -(2m+1)g_{m+1} - (m^2 - 4r^2)g_m + 8mr^2g_{m-1} + 4m(m-1)r^2g_{m-2}, \quad (51)$$

where terms with negative indices have zero coefficient and are omitted. Converting these factorial moments to raw moments with Stirling numbers, then raw moments to cumulants, checks (9) and higher orders independently.

For the Gaussian algebra, if $\exp(\sum_{h \geq 1} e_h v^h) = \sum_{k \geq 0} P_k v^k$, then

$$P_0 = 1, \quad P_k = \frac{1}{k} \sum_{h=1}^k h e_h P_{k-h}.$$

Take $e_h = \kappa_{h+2}(it)^{h+2}/(h+2)!$, adding ijt to e_1 for monomial amplitude z^j . Integrating each polynomial by Gaussian moments and multiplying the Stirling series gives the $H_\ell(j)$ of (28). Averaging over (25) reproduces the three displayed correction coefficients. This recurrence avoids merely repeating the tuple enumeration in (11).

10.3 Numerical diagnostics

For illustration let $B_n = a_n/(C_\epsilon(dn)^n)$ and $R_2(n) = n^3(B_n - 1 - c_{\epsilon,1}/n - c_{\epsilon,2}/n^2)$. Exact integer counts followed by high-precision evaluation give

n	$R_2(n)$	$n d_{TV}(J_n, J_\epsilon)$	first corrected $n^2 \ell^1/2$
40	-0.011103362959	0.1186222051	0.0468622340
80	-0.011235240759	0.1188802051	0.0471002563
160	-0.011304345034	0.1190101353	0.0472195093
41	-0.070685517647	0.1662254163	0.0765881923
81	-0.070899449934	0.1671477058	0.0766029313
161	-0.071006847771	0.1676175746	0.0766112777

The final column uses the signed approximation $\pi_\epsilon(j)(1 + Q_{\epsilon,1}(j)/n)$, so it is a half ℓ^1 error rather than a distance between two probability laws. The Poisson sums were numerically truncated beyond $n + 100$; they are diagnostics, not outward-rounded total-error certificates. The $R_2(n)$ values approach the two $c_{\epsilon,3}$ in Section 5. Subtracting that term gives stable n^4 -scaled residuals as well.

Separate bivariate counts at $n = 20, 21, 60, 61, 120, 121$ check the occupation mean offsets and characteristic functions. Marked evaluations at the cancellation points in Remark 4.2 test the necessity and correctness of the absolute leading formulation. These are additional numerical checks, not replacements for the complex-uniform proof.

[write] Rechecked, 7 October 2026. From its own exact counts (a rational power of F) the write reproduced every entry of the table above to the printed digits, with the Poisson sums carried to $j = n + 200$. From exact marked counts, $EK_n - np = -0.07093, -0.31307, -0.07109, -0.31237$ at $n = 60, 61, 120, 121$, against the offsets -0.071243 and -0.311664 of (33), with differences halving as n doubles. Observations, not tests of the asymptotic statements.

10.4 Package use and scope

The accompanying ZIP includes this L^AT_EX source, the PDF, standard-library exact verification, optional SymPy/mpmath diagnostics, recorded data, and a deterministic no-shell-escape builder. The core and optional dependencies are segregated. The detailed commands and finite test ranges are listed in `README.md` and `README_CODE.md`; the mathematical verifier can be run with Python’s `-S` flag, without site packages. The builder performs explicit runtime checks also under `-O`, rejects unresolved references and layout defects, and packages a complete SHA-256 inventory. Rebuilding the same sources with the same Python and TeX stack gives byte-identical artifacts; cross-version PDF identity is not promised.

The manifest is an integrity inventory, not a digital signature. The builder does not install dependencies or access the network, and refuses existing output paths. Optional high-precision diagnostics are not required for the exact core or PDF build. No third-party copyrighted book or article PDF is redistributed.

[write] **The shipped layout, 7 October 2026.** In ProveIt the “accompanying ZIP” is not shipped as such: the PDF and the checksum manifest are not shipped, the README is replaced by the report README, and the programs and records are shipped under `code/` and `data/` with prefixed names (Section 1.1); the delivered programs expect the delivered layout, and the report README gives a route that restores it on a copy.

11 Source comparison and limitations

The inspected official OEIS source record already contains the exact Bessel formula and Kotěšovec’s leading equivalent, including both numerical parity constants. Those results are not claimed as new here. The report supplies analytic definitions and a self-contained proof, then develops the stated finite-order, distributional, and inverse refinements. OEIS A328718’s fixed-dimensional row comment is not an unresolved problem being solved by this proportional-dimensional argument.

Gessel–Weinstein–Wilf’s primary article [3] was inspected in full. Its equation (2) gives the Bessel-product walk exponential generating function; its later permutation and Toeplitz-point results concern a different asymptotic framework. Flajolet–Sedgewick’s Theorem VIII.8, full proof, and subsequent vanishing-multiplier remarks [4] were read from the complete author-hosted book. They already provide the generic full-expansion and compact-ratio machinery. Period-two extraction, or the explicit two-arc proof here, handles our even support. We do not infer novelty from writing out that classical method.

A bounded comparison of the ProveIt report corpus at commit `83befe707f2840c2b53e0606701d8a2b28598e47` and of available earlier reports found no matching A328716 treatment in the inspected results. Exact identifier, decimal-constant, Bessel-formula, and lazy-walk searches were used. Nearby reports were actually inspected where relevant: the A039831 two-Fourier-peak maximum-coefficient problem differs from this bridge family, while proposed A047909 and A108242 directions were rejected as already covered. Report 169’s packed-matrix family was also excluded. These comparisons are bounded evidence, not an absence certificate. Earlier-report search missed known duplicates and therefore cannot certify novelty by a negative result.

Searches under Bessel-distribution and conditional-occupancy terminology also found general related literature without identifying this full family-specific package. That search was not exhaustive. Full historical priority for individual probability refinements remains unestablished. The accompanying source audit identifies the inspected sources and these limits. No OEIS record or external repository was edited or submitted as part of this report.

The mathematical limits are also explicit: no growing truncation order, boundary-ratio uniformity, fixed-dimensional asymptotic consequence, total-variation Gaussian limit, or effective numerical inverse threshold is asserted. The weighted discrete law, analytic occupation expansion, and eventual width-one inverse theorem hold precisely in the regimes proved above.

[write] **Added 7 October 2026.** The write read the A328716 entry itself (Remark 1.1); it agrees with the source’s account of the GitHub record, including both attributions and dates. The “fixed-dimensional row comment” of A328718 is stated there as a conjecture; Remark 9.2 proves it by the classical local limit argument, which is a fixed-dimension argument, so the source’s sentence that it is not “being solved by this proportional-dimensional argument” stays accurate.

12 Further questions within this treatment

Several concrete extensions remain outside the results proved here. These are questions for extending this treatment, not claims that the corresponding problems are historically unsolved.

1. Make the compact saddle bounds effective: produce explicit remainder constants and verified onsets, then combine them with interval evaluations of r , C_ϵ , and the inverse carriers. That would turn the existential brackets of Theorem 8.1 into a certified finite-input inverse calculator.
2. Analyze sparse and dense ratios $N/D \rightarrow 0$ and $N/D \rightarrow \infty$. The compact theorem does not control these limits; the saddle, variance, and relative sizes of the zero-step and coordinate occupancies require separate scaling arguments.
3. Strengthen the occupied-axis weak limit to a lattice local theorem or a joint Edgeworth expansion with the zero-step count. Such a result should explicitly retain the exact support and parity constraints, and specify the range of uniformity and tail norm rather than infer it from the characteristic-function limit alone.
4. Compare the finite-size polynomials in Theorem 6.1 and the occupation corrections systematically with general conditional-occupancy and Bessel-distribution literature. This could clarify alternative formulations and attribution beyond the bounded source search used here.

[write] **Added 7 October 2026, under Vladimir’s standing rule of 4 October 2026.** The four questions stand as the source states them. The write found no unproved claim of the source outside them and no claim of the source that is wrong; three printed constants are rounded rather than truncated in their last digit (note at the end of Section 3). On question 2: the extreme case of the dense ratio $N/D \rightarrow \infty$ in which D stays fixed is settled for its leading term by Remark 9.2; $N/D \rightarrow \infty$ with $D \rightarrow \infty$, and $N/D \rightarrow 0$, stay open.

References

- [1] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, A328716. Exact formula attributed to Ilya Gutkovskiy (26 October 2019); leading equivalent and numerical constants attributed to Václav Kotěšovec (27 October 2019). [Entry](#); [official source record](#). Inspected 3 October 2026.
- [2] OEIS Foundation Inc., A328718, square array of lazy closed lattice walks. Entry by Seiichi Manyama (26 October 2019); walk interpretation by Alois P. Heinz (26 October 2019). <https://oeis.org/A328718>. Inspected 3 October 2026.

- [3] I. Gessel, J. Weinstein, and H. S. Wilf, *Lattice walks in $\mathbb{Z}^d$ and permutations with no long ascending subsequences*, Electronic Journal of Combinatorics **5** (1998), Research Paper 2, 11 pp. [Primary journal PDF](#), especially equation (2), pp. 2–3.
- [4] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009. Theorem VIII.8 and proof, pp. 587–588; remarks and Note VIII.37, p. 589. [Author-hosted book PDF](#).